PT MERDEKA TSINGSHAN INDONESIA

STANDARD OPERATING PROCEDURE

ELECTRIC SINGLE BEAM CRANE

电动单梁起重机

MTI-CHP-CPM-SOP-012_Rev. 000

Information Security Classification
B – Business Information

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1. Purpose 目的

This procedure covers the operator's responsibility for visual inspection, start up and shutdown of the electric single beam crane under complete closure.

此程序涵盖操作员对完全关闭下的电动单梁起重机进行目视操作前检查、启动运行和停车的责任。

2. Functional Description 功能说明

It is the responsibility of the operator to ensure that the electric single beam crane is visually inspected before and after each shift to determine whether it can start and operate safely, or whether other activities (such as maintenance and repair or other operator tasks) must be performed before starting, to determine the sequential operation method of the electric single beam crane and smooth operation, Stop order of normal stops and emergency stops.

操作员有责任确保在从停机启动之前和每个班次开始时对电动单梁起重机进行目视检查，确定是否可以启动并安全运行，或者是否必须在开始启动之前执行其他活动(例如维护修理或其他操作员任务)，确定电动单梁起重机的按照顺序启动的操作方法及平稳运行，正常停车和紧急停车的停止顺序。

3. References 参考

- MTI-CP-SOP-Post Technical Operation Procedures. MTI-CP-SOP岗位操作规程
- Electric Single Beam Crane. 电动单梁起重机

4. PPE 个人保护装备



耳塞　安全帽　眼镜　劳保鞋　手套　工作服



防尘口罩

5. Hazards & Control 危害与控制

Description of Hazard / Environmental Issue 危害	Control Strategy 控制
1. Mechanical injury; 2. Electric shock; 3. Slippery conditions and a trip to the danger; 4. Noise;	1. Wear protective equipment according to the requirements, and no standing people or pedestrians under the sling; 2. Do not contact the electrical switch with wet hands during operation,Regular

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Description of Hazard / Environmental Issue 危害	Control Strategy 控制
5. Object falling from high altitudes. 1. 机械伤害； 2. 触电； 3. 湿滑的条件，绊倒危险； 4. 噪音； 5. 高空坠物。	inspection of the equipment leakage protection device is complete, effective; 3. Please hold the handrail up and down the stairs; 4. Wear the earplugs as required; 5. Wear labor protection goods and check the safety of stacking items on the operation platform. 1. 按要求穿戴好防护用品，吊物下严禁站人或行人； 2. 操作时不要湿手接触电器开关，定期检查设备漏电保护装置齐全、有效； 3. 上下楼梯请扶好扶手； 4. 按要求佩戴好耳塞； 5. 5.穿戴好劳保护品，检查各平台的物品堆放安全性。

6. Equipment 工具/ 设备

	Clean the site tools清扫现场工具

7. Procedure 程序

№	Stage	Key Points
1	General Check 普查	1. Confirm the pre-job technical training; 2. Wear suitable labor protection products, check and confirm that there is no unsafe state or safety risks on the site; 3. Check and clean the equipment and the site sanitation; 4. Contact with the process before and after, and give clear instructions. 1. 确认经过岗前技术培训; 2. 穿戴好适合的劳保护品，检查确认现场没有存在不安全状态或安全隐患; 3. 检查并打扫设备、现场卫生; 4. 与前后工序取得联系，并发出明确指令。
2	Inspect Electric Single Beam Crane 检查电动单梁起重机	1. Check the hook should be no cracks, serious deformation and wear; 2. Check the visible part of the wire rope should not exceed the standard, excessive wear and serious corrosion and other phenomena, and there should be no bending, flattening and other deformation, The wire

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№	Stage	Key Points
		rope lubricating oil is good, and the wire rope is arranged neatly on the drum; 3. Check whether the drum has cracks, excessive or abnormal wear, and the transverse moving momentum should not be too large; 4. Check the rope guide should effectively row the whole working range, and there should be no defects such as blockage and missing parts; 5. Check the emergency stop button of the control box to act reliably, and all function buttons work normally; 6. Check the limiter, lifting limiter and overspeed protection device are flexible and reliable; 7. Check that the slip line is straight, and it can slide freely and unimpeded. 1. 检查吊钩应无裂纹和严重变形、磨损； 2. 检查钢丝绳可见部分不应有断丝超标、磨损过量及严重腐蚀等现象，且不应有折弯、压扁等变形，钢丝绳润滑脂良好，钢丝绳在卷筒上排列整齐； 3. 检查卷筒是否有裂纹、过度或异常磨损，横向窜动量不应过大； 4. 检查导绳器应在整个工作范围内有效排绳，不应有卡阻、缺件等缺陷； 5. 检查操纵手柄紧急停止按钮动作可靠，各功能按钮工作正常； 6. 检查限位限制器、起重量限制器、超速保护装置灵活可靠； 7. 检查滑线平直，可自由无阻地滑动。
3	Inspect Materials Level 检查物料料位	1. Check whether the lifting plate or bucket can meet the material lifting capacity; 2. Check whether the material to be lifted is loaded or bucket, and whether the material is lower than the upper edge of the bucket; 3. Check whether the materials that do not need to be loaded or barrels are bound strong; 4. Check whether the material to be lifted is within the capacity of the crane. 1. 检查起重的装盘或桶是否能满足物料起重提升； 2. 检查需起重的物料是否装盘或桶，物料是否低于盘或桶上沿；

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№	Stage	Key Points
		3. 检查不需装盘或桶的物料是否捆绑结实； 4. 检查需起重提升的物料是否在起重机能力范围内。
4	Start-up of Electric Single Beam Crane 启动电动单梁起重机	Implement the driving "ten not hanging" system ⟹ Drive the power supply and start the handheld operation handle ⟹ lick the button to confirm that there is no abnormality ⟹ Lift the material with a hanging hook ⟹ Lift the material to the required location. 执行行车“十不吊”制度 ⟹ 驱动电源送电，启动手持操纵手柄 ⟹ 点动按钮确认无异常 ⟹ 将提升物料挂上吊钩 ⟹ 提升物料行进到必需的位置。
5	The Electric Single Beam Crane Stops 停止电动单梁起重机	1. Orderly shutdown Material is placed on the ground or on a smooth table ⟹ (2) After removing the hook on the lifting material, lift the empty hook to the upper limit of the hook that can not be touched by people or equipment ⟹ Open the electric hoist to the specified position ⟹ Turn off the electric gourd power supply ⟹ Hold the handheld control handle in the specified position ⟹ Clean the site for hygiene. 1. 正常停车 物料放在平整地面或平稳的台面 ⟹ 摘除提升物料上的吊钩后，将空吊钩提升到人或设备等碰不到的吊钩上限 ⟹ 把起重机空载开到指定位置有效锚定 ⟹ 关闭起重机电源 ⟹ 手持操纵手柄置于指定位置 ⟹ 打扫现场卫生。 2. Emergency Stop a. In an emergency, Check whether the start / stop button of the crane operating handle is stopped and turn off the drive power; b. After the accident, the equipment is restored to normal. If production still needs to be conducted, and the operation shall be restarted in the driving sequence; c. After the accident treatment, the equipment returns to normal. If the production is not necessary, operate according to the normal parking order. 2. 紧急停车 a. 当出现紧急事故，检查起重机操纵手柄的启动/

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№	Stage	Key Points
		停止按钮是否处于停止状态，关闭驱动电源； b. 事故处理后，设备恢复正常，如仍需要进行生产，按照开车顺序重新启动操作； c. 事故处理后，设备恢复正常，如不需要进行生产，按照正常停车顺序进行操作。
6	Permits & isolation sheets 许可证和隔离表	1. Only authorized isolators can sign off and cancel permits and registration sheets. 1. 只有经过授权的隔离者才能签署和取消许可证和登记表。

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8. Electric Single Beam Crane 电动单梁起重机 Questionnaire

Declaration of Understanding

I (_____Learner_____) have completed the theory assessment and had all my valid, reliable, and consistent with my level of knowledge in the assessment task.

Signature
Date:

I (_____Assessor_____) have witnessed the above learner complete the theory assessment and believe the assessment has provided a valid, reliable, and consistent result.

Signature
Date:

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9. Appendix

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Standard Operating Procedure
标准操作程序

Electric single beam crane in the chloride plant
CP 电动单梁起重机

SOP Details 程序明细

Processing Department 氯化厂	Procedure Level 程序级别: **3**	Revision Number 修订编号: 0
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